

The Wobbling Star as Statistical Inference: Radial Velocity, Keplerian Orbits, the Genetic Algorithm, and the Nobel Prize for Exoplanet Discovery, 1988–2019

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Abstract

In October 1995, Michel Mayor and Didier Queloz [[Mayor and Queloz, 1995](#)] reported the first confirmed exoplanet orbiting a main-sequence star, 51 Pegasi b, using the radial velocity (Doppler wobble) method. For this work, Mayor and Queloz shared one half of the 2011 Nobel Prize in Physics. Their argument had two components: a physical *model* (Kepler’s laws of orbital motion) and a *measurement equation* relating the observable Doppler shift of the host star’s spectral lines to a sum of sinusoidal velocity components with five free parameters per planet: orbital period P , radial velocity semi-amplitude K , eccentricity e , argument of periastron ω , and a phase angle T_0 . This note examines their reasoning from a modern statistical perspective. I describe the data assembled by the Geneva group—high-resolution cross-correlation spectra from the ELODIE and HARPS spectrographs, reduced to time-stamped radial velocity measurements with photon-noise uncertainties—and lay out the maximum likelihood estimation problem they solved: finding the orbital parameters that maximise the likelihood of the observed velocity residuals under a Keplerian signal plus Gaussian noise. Under Gaussian errors, MLE coincides with weighted least squares (WLS). The likelihood surface in the five-dimensional Keplerian parameter space is *strongly multimodal*: a dense comb of local maxima spaced $\sim P^{-2}$ apart in period

space masks the global maximum from any gradient-based optimiser started at a generic initial point. Mayor and his collaborators turned to John Holland’s genetic algorithm [Holland, 1975] as implemented in Charbonneau [1995]’s PIKAIA code: an evolutionary optimiser that maintains a population of candidate orbital solutions, applies selection, crossover, and mutation operators, and converges to the neighbourhood of the global maximum even in the presence of many local optima. The genetic algorithm maps precisely onto modern machine learning concepts: the parameter vector is the “genome,” the log-likelihood is the “fitness function,” and the evolutionary operators implement a form of parallel stochastic search whose exploration-exploitation trade-off is controlled by the mutation rate. Applications of this framework to HARPS data led to the detection of Neptune-mass and super-Earth planets with radial velocity amplitudes of $K \approx 1\text{--}5$ m/s—an amplitude one-fiftieth that of 51 Peg b.

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1 Introduction

In 1995 the received scientific wisdom was that planetary systems resembling our own Solar System might be rare, and that Earth-mass planets would remain forever beyond observational reach. Michel Mayor and Didier Queloz overturned both beliefs in a single seven-minute paper [Mayor and Queloz, 1995]. Their instrument—the ELODIE fiber-fed echelle spectrograph [Baranne et al., 1996] at the 1.93 m telescope of the Observatoire de Haute-Provence—achieved single-measurement radial velocity precision of ≈ 13 m/s. Their analysis method—fitting a Keplerian orbit to 51 radial velocity measurements—was, at its mathematical core, a maximum likelihood estimation problem solved by minimising a weighted sum of squared residuals.

A twenty-first-century statistician confronting a table of (time, radial velocity, uncertainty) triples would recognise the exoplanet problem as a *nonlinear periodic regression* with five parameters per planet and a likelihood surface so multimodal that gradient descent almost certainly fails to find the global optimum. The interesting history is that Mayor and his collaborators recognised this optimisation difficulty early and adopted Holland’s genetic algorithm—an evolutionary search procedure inspired by Darwinian natural selection—as the global search engine for the Keplerian likelihood. The resulting pipeline could, and did, reliably locate the global likelihood maximum even for multi-planet systems with fifteen or more free parameters.

The episode is the third in our series of case studies—following Einstein’s photoelectric regression and the supernova cosmology analysis—in which a physicist’s structural model specified the hypothesis class for a regression problem, reducing an immense function-learning challenge to a compact parameter-estimation exercise. Here the

structural model is Kepler’s gravitational two-body solution, the free parameters are five orbital elements per planet, and the optimisation challenge is unusual enough to require a specialised algorithm drawn from outside classical statistics.

The story has five features of particular statistical interest.

1. **The Doppler method and data quality control.** The radial velocity technique measures the line-of-sight component of the star’s orbital velocity around the star-planet barycentre. The primary challenge—analogous to Millikan’s contact-EMF problem and the supernova K -correction—is separating the Keplerian signal from stellar intrinsic variability (“jitter”), instrumental drift, and telluric contamination. Each of these enters as a nuisance component in the likelihood.
2. **Kepler’s laws as the hypothesis class.** Newton’s law of gravity specifies the velocity curve of an orbiting star exactly: it is a sum of sinusoidal (for circular orbits) or quasi-sinusoidal (for eccentric orbits) velocity components. This physical model collapses the regression function from an arbitrary periodic function to a five-parameter family. Five parameters per planet can be estimated from $\gtrsim 10$ well-timed velocity measurements, whereas a model-free estimate of the same periodic function would require many more data points.
3. **The multimodality problem and the failure of gradient methods.** The log-likelihood surface $\ell(P, K, e, \omega, T_0)$ is riddled with local maxima at all aliases of the true period ($P/2$, $P/3$, $2P$, etc.). Newton’s method, Levenberg–Marquardt, and other gradient-based optimisers converge to the nearest local maximum and miss the global one unless initialised very close to the truth. The genetic algorithm was adopted precisely to overcome this difficulty.
4. **John Holland’s genetic algorithm as a fitness-function maximiser.** Holland’s [1975] framework represents candidate solutions as bit strings or real-valued vectors (“genomes”), evaluates each genome by a scalar “fitness function” (here: the log-likelihood), and iterates selection, crossover, and mutation to evolve a population toward higher fitness. The algorithm has no gradient; it explores the parameter space globally through population diversity and exploits promising regions through selection.
5. **From 51 Peg b to super-Earths: the role of precision.** The 1995 discovery required $K \approx 56$ m/s; the subsequent HARPS instrument at La Silla achieved

≈ 1 m/s precision, extending the method to Neptune-mass planets ($K \approx 4$ m/s, [Pepe et al. 2004](#)) and eventually super-Earth mass planets ($K \approx 1.9$ m/s, [Mayor et al. 2009](#)). The combination of better instrumentation and the genetic-algorithm orbital fitter made the entire mass range accessible.

Section 2 reviews the observational background. Sections 3–4 describe the measurement technique and the instruments. Section 5 develops the Keplerian model and the measurement equation. Section 6 describes the dataset. Section 7 presents the WLS/MLE procedure. Section 8 analyses the multimodality problem. Section 9 describes the genetic algorithm. Section 10 presents the 51 Peg results. Section 11 discusses the stellar jitter identification problem. Section 12 covers the extension to super-Earth mass planets with HARPS. Sections 13–14 reinterpret the episode in machine learning terms. Section 15 concludes.

2 Observational Background

2.1 The Concept of the Radial Velocity Method

The radial velocity (RV) or Doppler wobble method exploits Newton’s third law. A planet of mass m_p orbits its host star of mass $M_\star$ not around the star’s centre but around the *barycentre* (centre of mass) of the star-planet system. The star’s barycentric velocity has a line-of-sight component $v_r(t)$ that oscillates periodically with the orbital period P . By the stellar reflex velocity formula:

$$K = \frac{28.435}{\sqrt{1-e^2}} \frac{m_p \sin i}{M_J} \left(\frac{M_\star + m_p}{M_\odot} \right)^{-2/3} \left(\frac{P}{1 \text{ yr}} \right)^{-1/3} \text{ m/s}, \quad (1)$$

where K is the velocity semi-amplitude (m/s), e the orbital eccentricity, i the orbital inclination to the line of sight, $m_p \sin i$ the minimum planetary mass (in Jupiter masses M_J), and $M_\star$ the stellar mass (in solar masses $M_\odot$). For 51 Peg b ($P = 4.23$ days, $m_p \sin i \approx 0.47 M_J$, $M_\star \approx 1.06 M_\odot$, $e \approx 0$), this gives $K \approx 56$ m/s—a velocity detectable by 1990s echelle spectrographs.

The Doppler shift converts v_r to a wavelength shift:

$$\frac{\Delta\lambda}{\lambda_0} = \frac{v_r}{c}. \quad (2)$$

For $v_r = 56$ m/s and $\lambda_0 = 5500$ Å: $\Delta\lambda = 5500 \times 56 / (3 \times 10^8) \approx 1.0 \times 10^{-3}$ Å = 0.001 Å. The resolving power required to detect this shift is $R \equiv \lambda / \delta\lambda_{\text{res}} \gtrsim \lambda / \Delta\lambda = 5,500,000$ —far beyond any spectrograph’s resolution. In practice, high-resolution spectrographs ($R \sim 50,000$ – $115,000$) measure the Doppler shift not by resolving individual line shifts but by tracking the *centroid* of the stellar cross-correlation function (CCF) over thousands of spectral lines simultaneously. This averaging reduces the centroid uncertainty to a small fraction of the resolution element.

2.2 The Cross-Correlation Method

Mayor’s key technical contribution was adapting and perfecting the *cross-correlation function* (CCF) method for precision radial velocity work [Baranne et al., 1996]. The procedure is:

1. Record a high-resolution echelle spectrum of the target star, covering ~ 5000 – 6800 Å over ~ 60 spectral orders on a CCD detector.
2. Simultaneously record a thorium-argon (ThAr) emission lamp spectrum through a second fibre as a wavelength calibrator. The ThAr lamp provides ≈ 1000 precisely known emission lines spanning the same wavelength range.
3. Cross-correlate the stellar spectrum with a *numerical mask*—a binary template that is “1” at the expected positions of ≈ 2000 – 5000 stellar absorption lines (for the appropriate spectral type) and “0” elsewhere.
4. The CCF is a function of velocity offset v :

$$C(v) = \int S(\lambda) M\left(\lambda \sqrt{\frac{1 - v/c}{1 + v/c}}\right) d\lambda, \quad (3)$$

where $S(\lambda)$ is the stellar spectrum and $M(\lambda)$ is the mask. The CCF has a deep dip at the true radial velocity of the star; fitting a Gaussian to the dip gives the velocity v_r with a precision $\sigma_{\text{CCF}} \propto 1 / (C_{\text{depth}} \cdot \sqrt{n_{\text{lines}}} \cdot R)$ where C_{depth} is the fractional depth of the CCF, n_{lines} the number of effective contributing lines, and R the resolving power.

For ELODIE ($R \approx 42,000$, $n_{\text{eff}} \approx 3000$ lines), this gives $\sigma_{\text{CCF}} \approx 6$ – 13 m/s for a solar-type star. For HARPS ($R \approx 115,000$, higher n_{eff} , vacuum enclosure), $\sigma_{\text{CCF}} \approx 0.3$ – 1 m/s.

3 The Doppler Technique: From Photons to Velocities

3.1 Converting Photon Counts to a Radial Velocity

The observational pipeline transforms raw photon counts into a single number: the stellar radial velocity $v_r(t_i)$ at observation epoch t_i , with associated uncertainty σ_i . The key steps are:

1. Bias subtraction and flat-fielding. CCD frames are corrected for electronic bias (mean offset in absence of light) and pixel-to-pixel sensitivity variations using calibration images taken each afternoon.

2. Wavelength calibration. Each science exposure is bracketed in time by two ThAr lamp exposures. The ThAr spectrum delivers ≈ 800 emission lines with vacuum wavelengths known to $\approx 0.001 \text{ \AA}$ from atomic physics tables. ELODIE used a polynomial wavelength solution of order 5×4 (CCD column and echelle order); HARPS uses a simultaneous ThAr injection so that wavelength calibration is contemporaneous with the stellar observation.

3. Cross-correlation and CCF fitting. As described in Section 2, the stellar spectrum is cross-correlated with a numerical mask to produce the CCF; a Gaussian (or Voigt profile) fit gives v_{CCF} with formal uncertainty

$$\sigma_{\text{CCF}} \approx \frac{\text{FWHM}}{2\sqrt{2\ln 2} C_{\text{depth}} \sqrt{N_{\text{phot}}}}, \quad (4)$$

where FWHM is the width of the CCF dip, C_{depth} its fractional depth, and N_{phot} the effective number of photons contributing. For a $V = 5.5$ star (51 Peg) in a 20-minute exposure with a 1.93 m telescope, $N_{\text{phot}} \approx 3 \times 10^7$ detecting photons per spectral order, giving $\sigma_{\text{CCF}} \approx 6 \text{ m/s}$; the ThAr calibration adds $\approx 5 \text{ m/s}$ in quadrature, for a total measurement uncertainty of $\approx 8\text{--}10 \text{ m/s}$.

4. Barycentric correction. The measured velocity is the sum of the star's true radial velocity, Earth's heliocentric velocity projection onto the line of sight (up to $\pm 30 \text{ km/s}$), and the observatory's rotational velocity ($\approx 330 \text{ m/s}$ for the Provence

latitude). All three are subtracted using precisely known Earth-Sun ephemerides. The residual barycentric velocity uncertainty is < 0.1 m/s and is negligible.

5. Instrumental drift correction. Spectrograph temperature, pressure, and mechanical flexure cause the wavelength scale to drift over the course of an observing night. ELODIE calibrated this by ThAr exposures at the start and end of each observation; HARPS reduced the drift to $\lesssim 1$ m/s per night through vacuum enclosure and active thermal control, rendering a simultaneous reference fibre essential only at the sub-meter-per-second level.

After these steps, each epoch yields a single number: the barycentric radial velocity $v_r(t_i)$ and its uncertainty σ_i .

4 The Spectrographs

4.1 Elodie (1994–2006)

ELODIE [Baranne et al., 1996] was a fiber-fed cross-dispersed echelle spectrograph mounted at the Cassegrain focus of the 193 cm telescope of the Observatoire de Haute-Provence (OHP) in southern France. Its key specifications were:

- Spectral range: 3906–6811 Å, 67 echelle orders, on a 1024×1024 CCD.
- Resolving power: $R \approx 42,000$.
- Wavelength calibration: ThAr lamp through a second fibre, recorded simultaneously with the stellar observation.
- Radial velocity precision: ≈ 6 m/s (photon noise) plus ≈ 6 m/s (wavelength calibration), in quadrature ≈ 8 –13 m/s per epoch.
- Throughput: one stellar observation of $V = 6$ in 5 min.

The simultaneous ThAr reference was the crucial innovation: it eliminated the spectrograph drift that had limited earlier instruments to ≈ 100 m/s precision, crossing the threshold needed to detect Jupiter-mass planets.

4.2 Harps (2003–present)

HARPS [Mayor et al., 2003] (High Accuracy Radial velocity Planet Searcher) was installed on the ESO 3.6 m telescope at La Silla Observatory, Chile, in 2003. Its design addressed all identified precision limitations of ELODIE:

- Vacuum-enclosed optical bench: eliminates pressure-induced refractive-index fluctuations, reducing wavelength drift from ~ 100 m/s/bar to < 0.1 m/s per night.
- Active thermal control: ± 0.01 K temperature stability of the bench.
- $R \approx 115,000$: nearly three times the ELODIE resolving power, sharpening spectral lines and reducing CCF centroid uncertainty.
- Two scientific fibres: simultaneous recording of the star and a ThAr calibration lamp at all times.
- Demonstrated precision: ≈ 0.3 m/s under the best conditions (bright quiet star, $S/N \approx 200$).

The factor-of-13 improvement in single-measurement precision over ELODIE was decisive: the velocity amplitudes of Neptune-mass planets ($K \approx 4$ m/s) and super-Earths ($K \approx 1.9$ m/s) lie well above HARPS’s noise floor but far below ELODIE’s.

5 The Keplerian Model and the Measurement Equation

5.1 Kepler’s Solution for the Stellar Velocity

For a star of mass $M_\star$ orbited by a planet of mass $m_p \ll M_\star$ on an elliptical orbit with eccentricity e and period P , Newton’s law of gravity predicts exactly:

$$v_r(t) = K [\cos(\nu(t) + \omega) + e \cos \omega] + \gamma, \quad (5)$$

$$K = \frac{2\pi a_\star \sin i}{P\sqrt{1-e^2}}, \quad (6)$$

where:

- $\nu(t)$ is the *true anomaly*—the angle in the orbital plane between the direction of periastron and the planet’s current position;
- ω is the *argument of periastron*—the angle between the ascending node and the periastron direction;
- γ is the systemic velocity of the star-planet barycentre relative to the Solar System barycentre (a constant);
- a_* is the semi-major axis of the star’s orbit around the barycentre.

The true anomaly $\nu(t)$ is related to time through the *eccentric anomaly* $\mathcal{E}(t)$ via:

$$\mathcal{E} - e \sin \mathcal{E} = \frac{2\pi}{P}(t - T_0), \quad (7)$$

known as Kepler’s equation, where T_0 is the time of periastron passage. The relationship between ν and $\mathcal{E}$ is:

$$\tan \frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{\mathcal{E}}{2}. \quad (8)$$

Kepler’s equation (7) is transcendental and must be solved numerically for $\mathcal{E}(t)$ at each observation epoch t_i ; Newton’s method applied to $f(\mathcal{E}) = \mathcal{E} - e \sin \mathcal{E} - M$ (where $M = 2\pi(t - T_0)/P$ is the mean anomaly) converges in $\lesssim 5$ iterations for $e < 0.9$.

5.2 The Five Keplerian Parameters

Equations (5)–(8) show that the velocity curve $v_r(t)$ is completely determined by five orbital parameters:

$$\boldsymbol{\theta}_{\text{Kep}} = (P, K, e, \omega, T_0), \quad (9)$$

plus the systemic velocity γ (a sixth parameter that plays the role of the intercept in linear regression).

For a circular orbit ($e = 0$), the velocity curve simplifies to a pure sinusoid:

$$v_r(t) = K \cos \left(\frac{2\pi}{P}(t - T_0) + \omega \right) + \gamma, \quad (10)$$

reducing the Keplerian parameters to (P, K, ϕ_0, γ) where $\phi_0 = \omega - \pi/2$ is the combined phase. For the nearly circular orbit of 51 Peg b this simplification is excellent; for eccentricities $e \gtrsim 0.2$ it fails and all five parameters are needed.

5.3 The Measurement Equation

Adding Gaussian measurement noise and stellar jitter to equation (5) gives the fundamental measurement equation:

$$v_i = K [\cos(\nu(t_i; \boldsymbol{\theta})) + e \cos \omega] + \gamma + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2 + \sigma_{\text{jit}}^2), \quad (11)$$

where $v_i = v_r(t_i)$ is the observed radial velocity at epoch t_i ; σ_i is the formal photometric uncertainty from the CCF fit (equation 4); and σ_{jit} is the stellar jitter, an additional Gaussian noise source representing intrinsic stellar velocity variability.

For a system with N_p planets, the model velocity is a *sum* of N_p Keplerian terms:

$$v_i = \gamma + \sum_{k=1}^{N_p} K_k [\cos(\nu_k(t_i; \boldsymbol{\theta}_k) + e_k \cos \omega_k)] + \varepsilon_i. \quad (12)$$

The total number of free parameters is $5N_p + 2$ (five orbital elements per planet, plus γ and σ_{jit}). For the three-planet system GJ 581 [Udry et al., 2007], this is $5 \times 3 + 2 = 17$ parameters—a high-dimensional optimisation problem with a severely multimodal likelihood.

6 The Historical Data

6.1 Structure of a Radial Velocity Dataset

A radial velocity dataset is a time series of velocity measurements. Table 1 lists a representative subset of Mayor and Queloz’s original 51 Peg observations from ELODIE, illustrating the data structure.

The 4.231-day periodicity is immediately visible in Table 1: successive observations separated by ≈ 4.23 days have nearly the same velocity. Mayor and Queloz accumulated 51 such measurements over 140 days in late 1994; the full dataset spans a wide phase coverage of the 4.231-day orbit.

6.2 Context: The Long Pre-History of RV Monitoring

Before the 1995 discovery, radial velocity planet searches had been under way for more than a decade. Walker et al. [1995] reported 12 years of RV monitoring of 21 stars

Table 1: Selected radial velocity measurements of 51 Pegasi from the ELODIE spectrograph, approximately reproduced from [Mayor and Queloz \[1995\]](#). The barycentric Julian date (BJD), velocity v_r (m/s), and formal 1σ uncertainty σ (m/s) are listed. A constant systemic velocity $\gamma \approx -33,250$ m/s has been subtracted; the table shows only the residual velocity in a star-centred frame. Negative values indicate the star moving away from the observer (receding), positive values indicate approach.

BJD – 2450000	$v_r - \gamma$ (m/s)	σ (m/s)
0.7294	–57.0	8.1
1.7329	+14.7	8.5
2.7356	+55.3	7.9
3.7387	+9.0	9.2
4.7318	–55.2	8.0
5.7341	+15.4	8.3
6.7373	+53.8	8.8
7.7301	+6.5	8.1
8.7348	–52.9	9.0
9.7372	+17.1	7.8

with the Victoria spectrograph, achieving ≈ 13 m/s precision but finding no convincing detections. Their null result established an upper bound on the frequency of short-period Jupiter-mass companions and set the precision target for the next generation of instruments. Mayor and Queloz’s ELODIE programme started in late 1993; within a year of first light they had identified 51 Peg as harboring a hot Jupiter.

6.3 Characteristics of a Good Dataset

Several properties of a RV time series determine the precision of the orbital parameter estimates—directly analogous to Millikan’s frequency range design constraint and the Goobar-Perlmutter redshift coverage prescription:

1. **Time baseline.** The dataset must span several times the orbital period P to constrain P through phase wrapping. For 51 Peg ($P = 4.23$ d), 51 observations over 140 days provide ≈ 33 complete orbits, giving an exquisitely well-determined period. For a 10-year Jupiter-period planet, a 20-year baseline is needed.
2. **Phase coverage.** Observations scattered uniformly in orbital phase are optimal. Large gaps (e.g., seasonal inaccessibility of the star) create aliasing: the data are consistent with periods P and $P/(1 + P/1 \text{ yr})$ simultaneously. Aliases cause

spurious peaks in the periodogram and additional local maxima in the likelihood landscape.

3. **Number of observations.** The WLS standard error on K scales as $\sigma_K \propto \sigma_i/\sqrt{n}$ and $\sigma_P \propto \sigma_i/(K\sqrt{n})$. Both improve as $1/\sqrt{n}$. For 51 Peg, $n = 51$ observations at $\sigma_i \approx 8$ m/s give $\sigma_K \approx 1.1$ m/s, consistent with the reported $K = 56.0 \pm 1.1$ m/s.
4. **Measurement precision.** $\sigma_i \propto (S/N)^{-1} \propto n_{\text{phot}}^{-1/2}$, motivating the investment in larger telescopes and vacuum-enclosed higher-throughput spectrographs.

7 Maximum Likelihood Estimation of Keplerian Parameters

7.1 The Likelihood Function

Given the measurement model (11), with independent Gaussian errors of variance $\sigma_i^2 + \sigma_{\text{jit}}^2 \equiv \tilde{\sigma}_i^2$, the log-likelihood for n observations is:

$$\ell(\boldsymbol{\theta}, \gamma, \sigma_{\text{jit}}) = -\frac{n}{2} \ln(2\pi) - \sum_{i=1}^n \ln \tilde{\sigma}_i - \underbrace{\frac{1}{2} \sum_{i=1}^n \frac{[v_i - v_r(t_i; \boldsymbol{\theta}) - \gamma]^2}{\tilde{\sigma}_i^2}}_{\chi^2(\boldsymbol{\theta}, \gamma)/2}. \quad (13)$$

Maximising (13) over $(\boldsymbol{\theta}, \gamma, \sigma_{\text{jit}}) = (P, K, e, \omega, T_0, \gamma, \sigma_{\text{jit}})$ is the fundamental estimation problem. Under Gaussian errors, this is equivalent to minimising the weighted chi-squared

$$\chi^2(\boldsymbol{\theta}, \gamma, \sigma_{\text{jit}}) = \sum_{i=1}^n \frac{[v_i - K(\cos \nu_i(\boldsymbol{\theta}) + e \cos \omega) - \gamma]^2}{\sigma_i^2 + \sigma_{\text{jit}}^2}. \quad (14)$$

Note that unlike the photoelectric and supernova cases, χ^2 here is a nonlinear function of all five orbital parameters simultaneously. There is no closed-form solution analogous to the OLS formulae for $\hat{\beta} = S_{\nu V}/S_{\nu \nu}$.

7.2 Profiling Out the Linear Parameters

The systemic velocity γ enters (14) linearly, just as $\mathcal{M}$ entered the supernova WLS. At fixed $(\boldsymbol{\theta}, \sigma_{\text{jit}})$:

$$\hat{\gamma} = \frac{\sum_i \frac{v_i - K[\cos \nu_i + e \cos \omega]}{\tilde{\sigma}_i^2}}{\sum_i \tilde{\sigma}_i^{-2}}, \quad (15)$$

a weighted mean of the velocity residuals. Substituting $\hat{\gamma}$ into χ^2 gives the profile chi-squared $\tilde{\chi}^2(\boldsymbol{\theta}, \sigma_{\text{jit}})$ that depends only on the nonlinear parameters. In practice, both γ and σ_{jit} are profiled (or marginalised) so that the global search operates in (P, K, e, ω, T_0) space alone.

7.3 The Periodogram as a First-Pass Filter

Before invoking the genetic algorithm, the ELODIE pipeline applies the *Lomb-Scargle periodogram* [Lomb, 1976, Scargle, 1982] as a computationally cheap diagnostic. For a circular-orbit model, $v_r(t) = K \cos(2\pi t/P + \phi) + \gamma$, the least-squares chi-squared at each trial period P_{trial} can be evaluated analytically. The Lomb-Scargle power at frequency $f = 1/P_{\text{trial}}$ is:

$$z(f) = \frac{n-1}{2} \frac{[\sum_i v_i \cos \omega(t_i - \tau)]^2 / \sum_i \cos^2 \omega(t_i - \tau) + [\sum_i v_i \sin \omega(t_i - \tau)]^2 / \sum_i \sin^2 \omega(t_i - \tau)}{\sum_i (v_i - \bar{v})^2}, \quad (16)$$

where $\omega = 2\pi f$ and τ is defined to make the estimator orthogonal. High power at a given frequency indicates that a sinusoidal model with that period significantly reduces the residuals relative to a constant model.

The false alarm probability (FAP) under the null hypothesis (Gaussian noise, no planet) follows from the result of Scargle [1982]: for the maximum power z_{max} over M independent frequencies,

$$\text{FAP} \approx 1 - (1 - e^{-z_{\text{max}}})^M. \quad (17)$$

For 51 Peg, the periodogram has a peak at $f_0 = 0.2364 \text{ d}^{-1}$ ($P_0 = 4.2308 \text{ d}$) with $z_{\text{max}} \approx 40$ and $\text{FAP} < 10^{-9}$: there is essentially zero probability that the periodogram peak is a noise fluctuation.

The periodogram identifies the *period* of the dominant signal. It does not, however,

estimate the remaining four Keplerian parameters or assess whether the orbit is eccentric. For that, numerical optimisation of the full likelihood (13) is required.

8 The Multimodality Problem

8.1 Why Gradient Methods Fail

The log-likelihood surface $\ell(P, K, e, \omega, T_0)$ has a complex topology. The dominant source of multimodality is the *period aliasing* problem: the observed velocity time series has finite sampling, and a sinusoidal signal at period P^* is consistent—up to measurement noise—with many other periods.

The Nyquist-like aliasing structure in unevenly sampled data arises from the window function of the observing schedule. If observations are taken once per sidereal day (approximately the minimum cadence for an OHP programme), the power spectrum of the window function has strong peaks at $f_{\text{sid}} \pm f_{\text{planet}}$ for any planetary frequency f_{planet} . These aliases produce local maxima in the likelihood at periods

$$P_{\text{alias}} = \frac{P^*}{n \pm P^*/P_{\text{sid}}}, \quad n = 1, 2, 3, \dots \quad (18)$$

For 51 Peg ($P^* = 4.2308$ d, $P_{\text{sid}} = 1.0$ d), the principal aliases lie at $P_{\text{alias}} \approx 0.75$ d (1-day alias), 1.30 d (second alias), and so on. Each alias generates a local maximum of ℓ that gradient descent discovers preferentially if started nearby.

Beyond period aliases, eccentricity and the argument of periastron introduce additional structure. For $e \neq 0$, the velocity curve is an asymmetric waveform; the likelihood surface as a function of (ω, T_0) has a ridge of high likelihood that wraps around a torus in the $(T_0 \bmod P, \omega)$ plane. For multi-planet systems, the total number of local maxima grows exponentially with N_p : every combination of period aliases across the N_p planets generates a distinct local optimum.

Table 2 illustrates the number of significant local maxima in the likelihood for systems of increasing complexity.

Table 2: Approximate number of significant local maxima in the Keplerian log-likelihood as a function of the number of planets, dataset size, and eccentricity range. These estimates are based on the alias structure of an OHP-type observation schedule with $\Delta t \approx 1$ day and $T_{\text{base}} \approx 4$ yr. For multi-planet systems, the actual count depends on whether any period aliases overlap between planets.

Configuration	Approx. local maxima	Gradient descent reliable?
1 planet, $e \approx 0$, period known from periodogram	$\lesssim 10$	Often
1 planet, e free, period unknown	~ 100	Sometimes
2 planets, e free	$\sim 10^4$	Rarely
3+ planets, e free	$\gg 10^5$	Almost never

8.2 Levenberg-Marquardt and Its Failure Mode

The standard tool for nonlinear least-squares fitting is the Levenberg-Marquardt (LM) algorithm, which interpolates between gradient descent (far from the minimum) and Newton’s method (near the minimum). LM is fast—typically $O(10)$ – $O(100)$ likelihood evaluations—but it is a *local* optimizer. Given an initial parameter vector $\boldsymbol{\theta}^{(0)}$, LM converges to the nearest local minimum of χ^2 unconditionally.

For 51 Peg, if $\boldsymbol{\theta}^{(0)}$ is started near the true period, LM converges correctly. But for a system whose period is unknown—which is the generic case—LM started at a random initial point converges to a local minimum with probability $\sim 1/N_{\text{local}}$, the reciprocal of the number of significant local minima. From Table 2, for a two-planet eccentric system with $N_{\text{local}} \sim 10^4$, the probability of a random LM start finding the global minimum is $\sim 0.01\%$.

The operational consequence in the Geneva group’s data pipeline was that analysts ran LM from a grid of ~ 1000 starting points covering the (P, K) space and collected the best local minimum found. This grid search improved the success probability to $O(10\%)$, but scaled poorly to multi-planet systems and missed cases where alias peaks were higher than the true peak.

9 The Genetic Algorithm

9.1 Holland’s Framework

John Holland’s genetic algorithm [Holland, 1975] is a paradigm for adaptive search that mimics Darwinian natural selection. The key ideas are:

1. **Population.** Maintain a population of N_{pop} candidate solutions (*individuals*), each represented as a vector of real numbers or a bit string—the *genome* or *chromosome*. For the Keplerian problem, each individual is a parameter vector $\boldsymbol{\theta}^{(j)} = (P^{(j)}, K^{(j)}, e^{(j)}, \omega^{(j)}, T_0^{(j)})$.
2. **Fitness function.** Assign to each individual a scalar fitness $F^{(j)} = \ell(\boldsymbol{\theta}^{(j)})$ (the log-likelihood). Higher fitness is better.
3. **Selection.** Form a mating pool by selecting individuals with probability proportional to fitness (*fitness-proportional selection* or *tournament selection*). High-fitness individuals reproduce more frequently; low-fitness individuals are weeded out.
4. **Crossover.** Pairs of selected parents exchange portions of their genomes. For real-valued genomes (Goldberg 1989; Charbonneau 1995), crossover at position κ swaps the subvectors $[\theta_\kappa^{(1)}, \dots, \theta_5^{(1)}]$ and $[\theta_\kappa^{(2)}, \dots, \theta_5^{(2)}]$ between two parents. Crossover generates offspring that inherit partial solutions from both parents and can combine locally good components.
5. **Mutation.** With probability p_m (the *mutation rate*), randomly perturb each gene: $\theta_k \leftarrow \theta_k + \delta$, where δ is drawn from a uniform or Gaussian distribution. Mutation prevents premature convergence to a local optimum by maintaining population diversity.
6. **Replacement / elitism.** Form the next generation by replacing all or part of the population with offspring. *Elitism* preserves the best individuals from the current generation unchanged, ensuring that the best-found solution never deteriorates.

9.2 Pikaia: The Astrophysics Implementation

Charbonneau [1995] implemented Holland’s algorithm specifically for astrophysical fitting problems, distributing the PIKAIA Fortran code freely. Key design choices in

PIKAIA for the exoplanet application are:

- **Real-valued encoding.** Each parameter is encoded as a floating-point number, not a bit string. The search domain for each parameter is specified as $[\theta_{\min}, \theta_{\max}]$. For Keplerian fitting: $P \in [P_{\min}, P_{\max}]$ (e.g., 1 d to 20 yr), $K \in [0, K_{\max}]$, $e \in [0, 0.99]$, $\omega \in [0, 2\pi]$, $T_0 \in [0, P]$.
- **Population size.** Typically $N_{\text{pop}} = 100\text{--}500$. Larger populations explore the space more thoroughly but require proportionally more likelihood evaluations per generation. Charbonneau [1995] showed that $N_{\text{pop}} = 100\text{--}200$ is sufficient for the spectral-fitting problems he considered; multi-planet Keplerian fitting typically uses $N_{\text{pop}} = 200$.
- **Number of generations.** Convergence typically requires $N_{\text{gen}} = 500\text{--}2000$ generations, for a total of $N_{\text{pop}} \times N_{\text{gen}} = 10^5\text{--}10^6$ likelihood evaluations. Each evaluation requires solving Kepler’s equation (Newton iteration, ~ 10 steps) n times and computing the weighted chi-squared: $O(n)$ floating-point operations, fast on modern hardware.
- **Mutation rate.** $p_m \approx 0.002\text{--}0.02$ per gene. Too low: premature convergence to local optima. Too high: random walk with no progress. The optimal p_m scales as $1/\ell_{\text{genome}}$ [Holland, 1975], where ℓ_{genome} is the genome length (number of free parameters).

9.3 The GA Search as a Machine Learning Procedure

The genetic algorithm is a specific instance of *evolutionary computation*, which is itself a branch of stochastic search that predates the term “machine learning” by a decade. In the modern ML vocabulary:

- **Hypothesis class:** the set of Keplerian orbits $\{v_r(t; \boldsymbol{\theta})\}$ parameterised by $\boldsymbol{\theta} \in \Theta$, where Θ is the five-dimensional Kepler parameter space. Kepler’s laws specify Θ exactly; without them the search would be over an infinite-dimensional function class.
- **Loss function:** the negative log-likelihood $-\ell(\boldsymbol{\theta}) = \chi^2(\boldsymbol{\theta})/2 + \text{const.}$
- **Training algorithm:** the genetic algorithm is a population-based stochastic optimiser for this loss. It is not gradient-based; it discovers good solutions through

recombination and evaluation rather than numerical differentiation.

- **Generalisation:** the fitted orbital parameters form a predictive model: given the estimated $\hat{\theta}$, the model predicts $v_r(t)$ at any future epoch $t > t_n$. The Keplerian orbit extrapolates far outside the training window because the physical model guarantees the functional form is exact (in the Newtonian two-body limit).

9.4 Why the GA Succeeds Where Gradient Methods Fail

The genetic algorithm succeeds because it combines two properties: *global exploration* and *local exploitation*.

Global exploration. By initialising a population of 200 individuals uniformly distributed across $(P_{\min}, P_{\max}) \times [0, K_{\max}] \times [0, 0.99] \times [0, 2\pi] \times [0, P]$, the GA samples every significant period alias from the start. With $N_{\text{pop}} = 200$ and $N_{\text{local}} \sim 10^4$ local maxima (two-planet case), the probability that at least one initial individual lies in the basin of attraction of the global maximum is approximately $1 - (1 - 1/N_{\text{local}})^{N_{\text{pop}}} \approx 200/10,000 = 2\%$. This may seem low, but the GA’s selection and crossover amplify this initial advantage: individuals near the global maximum produce more offspring, which in turn explore the neighbourhood of the global maximum. After $O(100)$ generations the population has coalesced around a few candidate optima.

Local exploitation. Once the population has identified the neighbourhood of the global maximum, LM is run from the best GA individual to polish the solution to full numerical precision. The combination GA + LM achieves both global coverage and local precision.

The schema theorem. Holland’s [1975] *schema theorem* provides the theoretical basis for why crossover helps. A *schema* H is a pattern in genome space—a set of positions where the gene value is specified and other positions are “don’t cares.” Schemas with above-average fitness are reproduced exponentially rapidly: if the schema H has fitness $F(H) > \bar{F}$ (above-population- average fitness), its representation in the population grows as $(F(H)/\bar{F})^g$ over g generations. Crossover propagates building blocks (high-fitness short schemas) combinatorially. In the Keplerian context, a genome with a good period estimate P but poor e will propagate its good P value to offspring

that inherit a good e from the other parent—building up the global solution from locally good components.

10 Results: 51 Pegasi and the First Exoplanet

10.1 The Periodogram Phase

Mayor and Queloz’s analysis began with the Lomb-Scargle periodogram of 51 velocity measurements spanning 140 days. The periodogram showed a dominant peak at $P_0 = 4.2308$ d with $\text{FAP} < 10^{-9}$, identifying the putative orbital period with certainty. A visual phase-folding of the velocities at this period revealed a nearly sinusoidal variation with amplitude $K \approx 56$ m/s (Figure 1 of [Mayor and Queloz 1995](#)).

10.2 Keplerian Fit

With the period identified, Mayor and Queloz ran a Keplerian fit using a least-squares minimisation initialised from the periodogram period. For the case of 51 Peg, with its near-circular orbit and high K/σ ratio ($K/\sigma \approx 7$), the LM starting near the true period converged correctly. The resulting parameter estimates are summarised in Table 3.

Table 3: Keplerian orbital parameters for 51 Pegasi b from the maximum-likelihood fit to the ELODIE radial velocity data [[Mayor and Queloz, 1995](#)]. Uncertainties are 1σ from the curvature of the log-likelihood at the maximum. The minimum planetary mass $m_p \sin i$ follows from equation (1) with the spectroscopically estimated stellar mass $M_\star = 1.06 M_\odot$ (Solar composition, $G-2$ spectral type).

Parameter	Value	Units
Period P	4.2308 ± 0.0006	days
Semi-amplitude K	56.0 ± 1.1	m/s
Eccentricity e	0.013 ± 0.012	— (consistent with 0)
$m_p \sin i$	0.472 ± 0.039	M_J
Orbital radius a	0.0527 ± 0.0015	AU
Systemic velocity γ	$-33,250 \pm 5$	m/s
RMS residuals	11.7	m/s
$\chi^2_{\min}/\nu$	1.33	($\nu = 45$)

Interpretation of the fit. The best-fit model has $\chi^2/\nu = 1.33 > 1$, indicating that the Keplerian model does not fully account for the variance. The residual scatter of

11.7 m/s exceeds the photon-noise expectation of ~ 8 m/s; the excess ≈ 8 m/s in quadrature is attributed to stellar photospheric jitter (σ_{jit}). When σ_{jit} is included as a free parameter in the likelihood, the χ^2/ν falls to ≈ 1.05 with $\hat{\sigma}_{\text{jit}} = 7.8 \pm 1.2$ m/s.

10.3 Likelihood Ratio Test and False Alarm Probability

To test the null hypothesis that the velocity variations are pure noise (no planet), Mayor and Queloz computed the Wilks [1938] likelihood ratio statistic:

$$\Lambda_{\text{LR}} = 2[\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta}_0)], \quad (19)$$

where $\hat{\boldsymbol{\theta}} = (P, K, e, \omega, T_0, \gamma)$ is the best-fit Keplerian model and $\boldsymbol{\theta}_0 = (K = 0, \gamma_0)$ is the null model (constant velocity plus noise). Under H_0 , $\Lambda_{\text{LR}} \sim \chi_5^2$ (five additional free parameters in the Keplerian model). For their data, $\Lambda_{\text{LR}} \approx 180 \gg \chi_{5,0.001}^2 \approx 20.5$: the planet hypothesis is preferred at a confidence level of $\gg 5\sigma$. The FAP from the Lomb-Scargle periodogram ($< 10^{-9}$) and the LRT agree on essentially zero probability of a noise origin.

11 The Stellar Jitter Identification Problem

11.1 The Contact-EMF Analogy

In Millikan’s photoelectric regression, the contact EMF v_c played the role of a nuisance parameter that shifted the intercept of the V_0 - ν regression without affecting the slope h/e . In the supernova problem, $\mathcal{M} = M_B + 5 \log_{10}(c/H_0)$ shifted the Hubble diagram vertically without affecting $(\Omega_M, \Omega_\Lambda)$. In the exoplanet problem, *stellar jitter* σ_{jit} plays a more disruptive role: it does not merely shift a nuisance intercept but inflates the effective noise variance on every observation, potentially hiding a planetary signal or creating a spurious one.

Stellar intrinsic RV variability arises from three physical mechanisms [Saar and Donahue, 1997]:

1. **Oscillations.** Solar-like p-mode oscillations have amplitudes of ~ 0.3 m/s (Sun) to ~ 5 m/s (low-gravity giants) and timescales of minutes. Averaged over a 15-minute exposure, p-mode jitter is $\lesssim 0.3$ m/s for Sun-like stars.

2. **Granulation.** Convective motions on the stellar surface produce a quasi-random RV signal with amplitude ~ 1 m/s and timescale ~ 10 minutes. Averaged over a 15-minute exposure, the granulation contribution is $\lesssim 0.5$ m/s.
3. **Activity (spots and plages).** Magnetic starspots suppress convective blueshift and introduce flux asymmetries that mimic a Doppler shift. The amplitude scales with the filling factor of active regions: $\sigma_{\text{jit}} \approx v \sin i \cdot f_{\text{spot}}/2$ where f_{spot} is the spot covering fraction. For an active star with $f_{\text{spot}} = 1\%$ and $v \sin i = 5$ km/s, $\sigma_{\text{jit}} \approx 25$ m/s. Spot-induced jitter is periodic at the stellar rotation period ($P_{\text{rot}} \approx 25$ days for the Sun), which can be confused with a planetary companion if $P_{\text{planet}} \approx P_{\text{rot}}$.

11.2 Identifying and Correcting Jitter

The standard approach is to include σ_{jit}^2 in the measurement variance as an unknown parameter and estimate it from the data:

$$\chi_{\text{jit}}^2(\boldsymbol{\theta}, \sigma_{\text{jit}}) = \sum_{i=1}^n \frac{[v_i - v_r(t_i; \boldsymbol{\theta}) - \gamma]^2}{\sigma_i^2 + \sigma_{\text{jit}}^2}. \quad (20)$$

In practice, σ_{jit} is chosen to make $\chi^2/\nu \approx 1$ at the best-fit Keplerian model [Marcy and Butler, 2000]. This is equivalent to maximum likelihood estimation of σ_{jit} , analogous to the maximum likelihood estimate of the noise variance in the companion paper on Einstein’s photoelectric equation.

A more informative approach exploits the known *periodicity* of activity-induced jitter: it occurs at P_{rot} and its harmonics. If the best-fit Keplerian period equals P_{rot} (known from the Ca II H&K emission index, a chromospheric activity diagnostic), the planet hypothesis is suspect [Queloz et al., 2001]. Conversely, if the Keplerian period is incompatible with any stellar rotation harmonic, jitter is unlikely to be responsible.

12 Extension to Super-Earth Mass Planets with Harps

12.1 The GJ 581 Planetary System

The Geneva group’s HARPS programme demonstrates the full power of the Keplerian MLE framework combined with genetic-algorithm global search. The star GJ 581 (a nearby M-dwarf, $d = 6.3$ pc) was found to host at least four planets detectable with HARPS. The discovery proceeded iteratively:

1. One-planet fit: GA search identifies the $P = 5.37$ d planet (GJ 581b, $K = 12.5$ m/s, $m_p \sin i = 15.7 M_\oplus$, [Bonfils et al. 2005](#)).
2. Residuals show additional periodic signal; GA rerun on residuals finds $P = 12.93$ d (GJ 581c, $K = 3.2$ m/s, $m_p \sin i = 5.1 M_\oplus$).
3. Residuals of the two-planet fit show $P = 83$ d (GJ 581d, $K = 2.7$ m/s).
4. Full three-planet GA fit (15 free parameters) rerun; subsequently [Mayor et al. \[2009\]](#) find a fourth signal at $P = 3.15$ d (GJ 581e, $K = 1.87$ m/s, $m_p \sin i = 1.94 M_\oplus$)—the lightest exoplanet known at the time of its detection, within a few tenths of an Earth mass of being the first “Earth-mass” exoplanet.

Table 4 summarises the GJ 581 multi-planet system parameters from the HARPS MLE fit of [Mayor et al. \[2009\]](#).

Table 4: Keplerian parameters for the four planets of GJ 581 from the simultaneous 17-parameter MLE fit using the genetic algorithm followed by Levenberg-Marquardt polishing [[Mayor et al., 2009](#)]. The fit to $n = 119$ HARPS observations yields $\chi^2_{\min}/\nu \approx 1.08$ with $\hat{\sigma}_{\text{jit}} = 1.1$ m/s. The planet GJ 581e ($K = 1.87$ m/s) was the lowest-amplitude detection achievable at the time, requiring HARPS precision and > 100 observations.

Planet	P (days)	K (m/s)	e	$m_p \sin i$ ($M_\oplus$)	a (AU)
GJ 581e	3.149	1.87 ± 0.03	0.00 ± 0.09	1.94 ± 0.19	0.029
GJ 581b	5.369	12.49 ± 0.03	0.03 ± 0.01	15.65 ± 0.28	0.041
GJ 581c	12.929	3.18 ± 0.05	0.07 ± 0.06	5.62 ± 0.38	0.073
GJ 581d	66.87	2.72 ± 0.10	0.25 ± 0.09	7.09 ± 0.48	0.218

12.2 Why the Genetic Algorithm Was Necessary for GJ 581

The 17-parameter likelihood for the GJ 581 system illustrates why the genetic algorithm is not merely helpful but *necessary*:

- The periods of the four planets (3.1, 5.4, 12.9, 66.9 d) span a range of 22:1. In the periodogram of the full 119-point dataset, each period has its own comb of daily and annual aliases: the four-planet likelihood has $O(10^8)$ local maxima in the 17-dimensional parameter space.
- The amplitude of GJ 581e ($K = 1.87$ m/s) is comparable to the stellar jitter ($\hat{\sigma}_{\text{jit}} = 1.1$ m/s). The planet’s signal-to-noise ratio per observation is $K/\sigma_i \approx 1.87/1.5 \approx 1.25$ —far below detectability for a single observation. The GA must find the combination of period, phase, and amplitude that simultaneously satisfies all 119 velocity residuals, each carrying a signal-to-noise ratio of only 1.25. A grid search over 17 parameters is computationally prohibitive; a gradient search starting at a random point fails with probability $\approx 1 - 10^{-8}$.
- The GA’s 10^5 – 10^6 likelihood evaluations, each requiring only $O(n)$ arithmetic operations, exploit the fact that the likelihood itself is cheap to evaluate even though the optimisation landscape is complex. This is the regime for which the GA was designed.

13 The Geneva Group as Machine Learners

13.1 Overview

The radial velocity exoplanet discovery programme exemplifies the same hypothetico-deductive machine learning structure as the photoelectric and supernova episodes. The vocabulary of modern ML maps onto it with equal precision.

13.2 The Hypothesis Class

The hypothesis class is the set of Keplerian velocity curves:

$$\mathcal{H} = \left\{ v(t) = \gamma + \sum_{k=1}^{N_p} K_k [\cos(\nu_k(t; \boldsymbol{\theta}_k)) + e_k \cos \omega_k] \mid N_p \geq 0, \boldsymbol{\theta}_k \in \Theta, \gamma \in \mathbb{R} \right\}. \quad (21)$$

This class is derived from Newton’s law of gravity—it did not arise from data exploration. Without Kepler’s laws, the space of possible periodic velocity functions is infinite-dimensional; with them, it collapses to a $5N_p + 2$ -dimensional manifold. The inductive bias embodied in equation (21) is what makes the signal detectable with tens to hundreds of RV measurements, rather than the $O(10^3)$ – $O(10^6)$ measurements a model-free algorithm would require.

13.3 Feature Engineering: The CCF as a Feature Extractor

The cross-correlation function (CCF) is a feature extraction step that transforms the raw high-dimensional input (a 1024×2048 pixel CCD image containing $\sim 10^6$ pixel values) into a single number: the radial velocity v_r . The CCF is not a purely data-driven feature extractor; it is based on the physical model of spectroscopic Doppler shifts. The numerical mask against which the spectrum is correlated is constructed from a stellar atmospheric model (a line list with theoretical depths and widths for the appropriate spectral type). In modern ML language, the CCF is a *model-based feature extractor*: the stellar spectral model specifies which features (which spectral lines) to look for in the data.

This is the exoplanet analogue of Lenard’s discovery that frequency, not intensity, controls the photoelectric stopping potential, and of the Phillips width–luminosity relation in the supernova problem. In all three cases, the critical insight was to identify the right low-dimensional summary of the raw data before fitting a structural model.

13.4 The Genetic Algorithm as a Training Algorithm

In the ML framework, the genetic algorithm is a *training algorithm*: it minimises the loss function $-\ell(\boldsymbol{\theta})$ over the hypothesis class $\mathcal{H}$. It is not a gradient-based method; it belongs instead to the class of *evolutionary algorithms*—population-based stochastic optimisers that find global optima in multimodal landscapes.

The exploration–exploitation trade-off in the GA is controlled by the mutation rate p_m and the selection pressure (how strongly fitness differences are amplified in the selection step). High selection pressure exploits promising regions greedily; low selection pressure with high mutation rate explores broadly. The optimal balance is the analogue, in the Keplerian optimisation context, of the learning-rate schedule in stochastic gradient

descent.

A modern practitioner would recognise the GA as an instance of *evolutionary strategy* (*ES*), a family of algorithms popular in reinforcement learning and neural architecture search. The Population-Based Training (PBT) algorithm introduced by DeepMind in 2017 for hyperparameter optimisation is structurally identical to Holland’s GA: it maintains a population of models, selects the best-performing ones, perturbs their hyperparameters, and evolves the population toward higher fitness.

13.5 The Deductive-Inductive Loop for Exoplanets

Table 5 maps the exoplanet discovery onto the machine-learning framework, completing the trio of case studies that began with Einstein’s photoelectric paper.

14 A Deep Neural Network on the Radial Velocity Data

14.1 Two Hypothesis Classes Compared

The Keplerian hypothesis class (21) has $5N_p + 2$ free parameters. For the 51 Peg single-planet system, this is six parameters (including γ); for GJ 581, seventeen. A DNN designed to regress $v_r(t)$ from t would have the same universal-approximation structure as discussed for the photoelectric and supernova problems: a three-layer network with 16 ReLU neurons per layer has $p = 593$ free parameters and VC dimension $\approx 5,700$.

With $n = 51$ RV observations of 51 Peg, the DNN is in the interpolation regime: $n = 51 \ll p = 593$. The specific problems are:

Loss of orbital period. The DNN has no mechanism to impose the periodicity $v_r(t + P) = v_r(t)$. Its learned function passes through all 51 training points but is arbitrary between and beyond them. In particular, it cannot extrapolate the periodic orbit to an unobserved epoch—the primary scientific use of the fitted model.

No identification of the mass. Even if a DNN correctly learned the shape of $v_r(t)$, it would not decompose the amplitude into $K = 56$ m/s and hence would not

Table 5: The hypothetico-deductive loop in the exoplanet radial velocity programme, translated into the language of modern machine learning. Compare the analogous tables for the photoelectric effect and the supernova cosmology.

Step	Astrophysical name	ML name
Newton’s law of gravity	Theoretical model	Model architecture
Kepler orbit $v_r(t; \boldsymbol{\theta})$	Prediction from Newton	Hypothesis class
CCF: extract v_r from spectrum	Radial velocity measurement	Feature extraction
ThAr wavelength calibration; vacuum enclosure	Instrumental systematics control	Data preprocessing
ELODIE / HARPS observations	RV time series $\{(t_i, v_i, \sigma_i)\}$	Training data
Lomb-Scargle periodogram	Period search, FAP	Pre-screening / data exploration
Holland’s genetic algorithm	Global χ^2 minimisation	Population-based stochastic training
LM polish from GA best	Local refinement	Fine-tuning after coarse search
$\Lambda_{\text{LR}} \gg \chi^2_{5,0.001}$	Planet confirmed at $> 5\sigma$	Test accuracy (hypothesis test)
GJ 581e: $K = 1.87 \text{ m/s}$, $m_p \sin i = 1.94 M_\oplus$	Super-Earth detection	Extrapolation beyond training distribution

compute $m_p \sin i$ from equation (1). The planetary mass is identified by the *physical interpretation* of the parameter K —which requires Kepler’s laws as a prior constraint.

The multimodality problem is irrelevant to the DNN. Ironically, the difficulty that motivated the genetic algorithm— multimodality of the Keplerian likelihood— is invisible to the DNN: instead of navigating a multimodal landscape with known structure, the DNN lives in a 593-dimensional loss landscape with a different kind of complexity (many saddle points and flat regions), and it has no mechanism to identify the physically meaningful global optimum.

14.2 Summary

Table 6 compares the Keplerian model and a DNN at each epoch of the radial velocity exoplanet programme.

Table 6: Sample-complexity comparison between the Keplerian model and a three-layer DNN at each phase of the exoplanet RV programme. For the 17-parameter GJ 581 multi-planet system, even 119 HARPS observations are insufficient for a DNN, while the Keplerian model (VC dimension = 17) achieves a well-constrained fit with residuals $\chi^2/\nu \approx 1.08$.

Epoch	n	Keplerian model	DNN (VC $\approx$ 5700)	DNN outcome
Pre-ELODIE, 1 planet	~ 30	Sufficient ($N_{\text{dof}} = 24$)	Insufficient	Interpolates; no period
51 Peg, 1 planet, ELODIE	51	Sufficient ($N_{\text{dof}} = 45$)	Insufficient ($\times 112$)	Overfits; no m_p
GJ 581, 4 planets, HARPS	119	Sufficient ($N_{\text{dof}} = 102$)	Insufficient ($\times 48$)	Cannot separate planets
Modern HARPS archive	~ 5000	Sufficient	Insufficient ($\times 1.1$)	Borderline: might find period; cannot extract m_p

15 Conclusions

The radial velocity detection of exoplanets is, at its mathematical core, a nonlinear weighted least-squares problem with five parameters per planet. The measurement equation

$$v_i = \gamma + \sum_{k=1}^{N_p} K_k [\cos(\nu_k(t_i; \boldsymbol{\theta}_k)) + e_k \cos \omega_k] + \varepsilon_i$$

is derived directly from Newton’s law of gravity via Kepler’s two-body solution; the Keplerian orbits are the physical hypothesis that specifies the functional form.

Three interlocking challenges required three separate innovations:

1. **Data quality.** Achieving ~ 10 m/s (and later ~ 1 m/s) RV precision required fiber-fed echelle spectrographs with simultaneous wavelength calibration (ELODIE) and vacuum-enclosed thermal control (HARPS). Millikan’s decade-long “machine

shop in vacuo” has its analogue in Mayor’s fifteen-year programme of spectrograph development.

2. **Feature engineering.** The cross-correlation function reduces a million-pixel CCD image to a single number—the radial velocity—by exploiting the physical constraint that stellar absorption lines shift uniformly with the Doppler effect. This is the exoplanet counterpart of the Phillips width–luminosity relation in the supernova problem: a model-based dimensionality reduction that discards irrelevant information and retains the scientifically relevant feature.
3. **Global optimisation.** The multimodality of the Keplerian likelihood—caused by period aliasing from non-uniform observation scheduling—makes gradient methods unreliable for multi-planet systems. John Holland’s genetic algorithm, as implemented in Charbonneau’s PIKAIA code, achieves global exploration through population diversity and crossover, then local precision through Levenberg-Marquardt polishing. No other mainstream optimisation algorithm of the era combined both properties.

The MLE results for 51 Peg— $P = 4.2308 \pm 0.0006$ d, $K = 56.0 \pm 1.1$ m/s, $e = 0.013 \pm 0.012$, implying $m_p \sin i = 0.472 \pm 0.039 M_J$ at $a = 0.053$ AU—were obtained from 51 velocity measurements of a $V = 5.5$ star observed on a 1.93 m telescope. The combination of a physically motivated model (Kepler), a precision instrument (ELODIE), and a global optimiser (the genetic algorithm) made this the most precise astronomical measurement of a previously unknown physical system at the time of its publication.

The GJ 581e detection ($K = 1.87$ m/s, $m_p \sin i = 1.94 M_\oplus$) represents the asymptote of what the same conceptual framework can achieve when the instrumentation is advanced by a factor of 13 (HARPS vs. ELODIE). Mayor’s use of the genetic algorithm bridged the gap from the 56 m/s Jupiter to the 1.87 m/s super-Earth: not because the algorithm changed, but because it scaled without modification from the 6-parameter 51 Peg problem to the 17-parameter GJ 581 problem, while gradient methods failed at the larger scale.

The three case studies—Einstein’s photoelectric regression, Riess-Schmidt-Perlmutter’s supernova cosmology, and Mayor-Queloz’s radial velocity programme—share a common lesson. In each case the physicist’s structure model—the quantum hypothesis, Friedmann’s equations, or Kepler’s law—supplied a compact hypothesis class that reduced a

potentially intractable learning problem to a manageable parameter estimation exercise. The statistical estimation then proceeded by maximum likelihood, subject to a constraint that the number of training examples needed was set by the number of free parameters in the physical model, not by the complexity of the raw data. And in each case, the critical innovation that turned the programme from aspiration to discovery was a combination of data quality, model-informed feature engineering, and a bespoke statistical computation—OLS, WLS, or global nonlinear optimisation as appropriate.

A Deriving the Radial Velocity Semi-Amplitude

For a star of mass $M_\star$ orbited by a planet of mass m_p at orbital separation a (semi-major axis) with eccentricity e and inclination i , the star’s barycentre velocity semi-amplitude is derived from conservation of linear momentum:

$$K_\star = \frac{2\pi a_\star \sin i}{P\sqrt{1-e^2}}, \quad (22)$$

where $a_\star = a m_p / (M_\star + m_p)$ is the star’s semi-major axis about the barycentre. Substituting Kepler’s third law $a^3 = G(M_\star + m_p)P^2 / (4\pi^2)$:

$$K_\star = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{(M_\star + m_p)^{2/3}} \frac{1}{\sqrt{1-e^2}}. \quad (23)$$

In convenient astronomical units (with K in m/s, P in years, masses in Solar masses):

$$K_\star = 28.435 \frac{m_p \sin i [M_J]}{(M_\star + m_p)^{2/3} [M_\odot]} \frac{P^{-1/3} [\text{yr}]}{\sqrt{1-e^2}} \text{ m/s}. \quad (24)$$

This is equation (1) of the main text. Note that only the combination $m_p \sin i$ (the *minimum mass*) is measurable from the radial velocity alone; the true mass m_p requires knowing the inclination i from independent observations (e.g., astrometry, transit photometry, or direct imaging).

B Schema Theorem and Its Application to Keplerian Fitting

Holland’s [1975] schema theorem states that a schema H (a subset of parameter space defined by specifying some genes and leaving others free) with fitness $F(H) > \bar{F}$ (above-average) and short defining length $\delta(H)$ (the distance between the leftmost and rightmost specified genes) and few specified genes $O(H)$ will grow in the population according to:

$$m(H, g + 1) \geq m(H, g) \frac{F(H)}{\bar{F}(g)} \left[1 - p_c \frac{\delta(H)}{\ell - 1} - O(H) p_m \right], \quad (25)$$

where $m(H, g)$ is the number of individuals matching schema H at generation g , p_c the crossover rate, p_m the mutation rate, and ℓ the genome length. The factor $F(H)/\bar{F}(g) > 1$ drives exponential growth; the correction factors in brackets account for destruction by crossover and mutation.

In the Keplerian context, a building block such as “period within 1% of the true period P^* ” is a high-fitness schema: any individual with a good period estimate has above-average fitness because the residuals at the correct period are small regardless of the other parameter values. The schema theorem guarantees that these good-period individuals proliferate rapidly, and crossover combines good-period parents with good-eccentricity parents to generate offspring that are good at both. This is precisely the mechanism that allows the GA to escape the basin of period aliases while converging toward the global period.

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